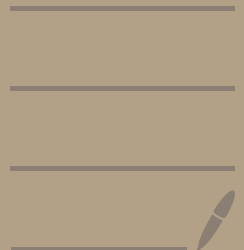


E, T17 Tipler



Escala de Temperatura

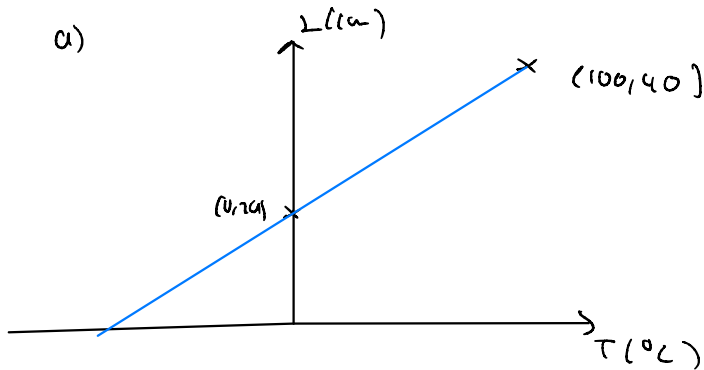
32. $T_F = \frac{9}{5} T_C + 32$

$$T_{F_2} = \frac{9}{5} (-12) + 32 = 10,4^\circ \text{F} \quad T_{F_2} = \frac{9}{5} (-77) + 32 = 19,4^\circ \text{F}$$

33. $T_C = \frac{5}{9} (T_F - 32) = \frac{5}{9} (1945,9 - 32) = 1063^\circ \text{C}$

34. $T_F = \frac{9}{5} \cdot 15 + 32 = 59^\circ \text{C}$

35.



b) $L_{22} = \frac{L_{22} - L_0}{L_{100} - L_0} \times 100 \Rightarrow 22 = \frac{L_{22} - 24}{40 - 24} \times 100$

$$L_{22} = 27,52 \text{ cm}$$

c) $t_c = \frac{L_t - L_0}{L_{100} - L_0} \times 100 = \frac{25,4 - 24}{16} \times 100 = 8,75^\circ \text{C}$

36) $1 \cdot 10^7 \text{ K} = 1 \cdot 10^7 - 273,15^\circ \text{C} = 9999726,85^\circ \text{C}$

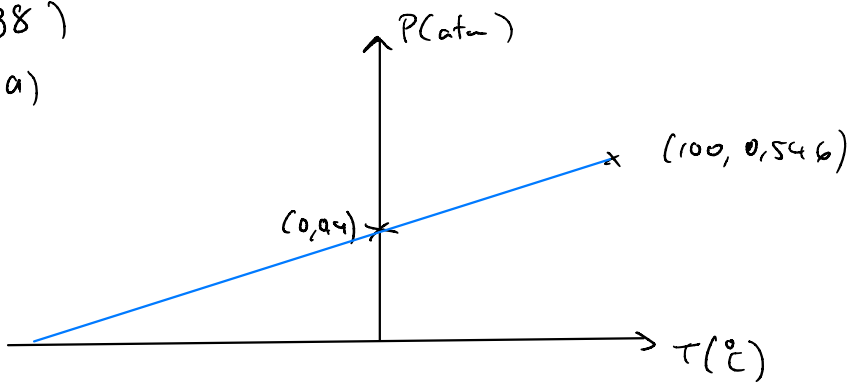
$$T_F = \frac{9}{5} T_C + 32 = 17999540,33^\circ \text{F}$$

37) $T_F = \frac{9}{5} (T_C) + 32 = \frac{9}{5} (T_K - 273,15) + 32$

$$T_F = -320,44^\circ \text{F}$$

38)

a)



b)

$$t_c = \frac{P_t - P_c}{P_{100} - P_0} \times 100 = \frac{0.1 - 0.4}{0.546 - 0.4} \times 100 = -205.5^\circ\text{C}$$

c)

$$t_{444,6} = \frac{P_{444,6} - P_0}{P_{100} - P_0} \times 100^\circ\text{C} \Rightarrow 444.6 = \frac{P_{444,6} - 0.4}{0.146} \times 100$$

$$P_{444,6} = 1.049 \text{ atm}$$

Ley de Los Gases Ideales

$$4b) 1 \text{ torr} = 133,3 \text{ Pa}$$

$$1 \cdot 10^{-4} \text{ torr} = 133,3 \cdot 10^{-8} \text{ Pa}$$

$$PV = NkT \rightarrow N = \frac{133,3 \cdot 10^{-4} \cdot 1 \cdot 10^{-6}}{1,38 \cdot 10^{-23} \cdot 300} = 3,22 \cdot 10^8 \text{ molecules}$$

$$50) T_0 = -8^\circ\text{C} = 265,15 \text{ K} \quad P_0 = 180 \cdot 10^3 \text{ Pa}$$

$$P_1 = 245 \cdot 10^3 \text{ Pa}$$

$$a) \frac{P_1}{T_1} = \frac{P_2}{T_2} \rightarrow \frac{180 \cdot 10^3}{265,15} = \frac{245 \cdot 10^3}{T_2}$$

$$T_2 = \frac{245 \cdot 10^3 \cdot 265,15}{180 \cdot 10^3} = \frac{49}{36} T_1 = 360,19 \text{ K}$$

$$b) \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \rightarrow V_2 = 1,07 V_1$$

$$\frac{180 \cdot 10^3 \cdot V_1}{T_1} = \frac{245 \cdot 10^3 \cdot 1,07 V_1}{T_2} \rightarrow T_2 = \frac{245 \cdot 10^3 \cdot 1,07}{180 \cdot 10^3} T_1$$

$$T_2 = 386,16 \text{ K}$$

$$51) V = 6 \times 5 \times 3 = 90 \text{ m}^3$$

$$a) PV = nRT \quad P = 1 \text{ atm} = 1,013 \cdot 10^5 \text{ Pa}$$

$$n = \frac{90 \cdot 1,013 \cdot 10^5}{8,314 \cdot 300} = 3655,28 \text{ mol}$$

$$b) n = \frac{90 \cdot 1,013 \cdot 10^5}{8,314 \cdot 305} = 3595,35 \text{ mol}$$

$$3655,28 - 3595,35 \text{ mol} = 59,92 \text{ mol salen}$$

$$52) m = 10g \quad T_0 = 4,2 K \quad P = 1 \text{ atm} = 1,013 \cdot 10^5 \text{ Pa}$$

$$n = \frac{m}{M} = \frac{10}{4,008} = 2,499 \text{ mol} \rightarrow n = 2,5 \text{ mol}$$

$$pV = nRT$$

$$1,013 \cdot 10^5 \cdot V = 2,5 \cdot 8,314 \cdot 4,2 \rightarrow V = 8,61 \cdot 10^{-4} \text{ m}^3$$

$$a) \frac{V_0}{T_0} = \frac{V_1}{T_1} \rightarrow V_1 = \frac{8,61 \cdot 10^{-4}}{4,2} \cdot 25 K = 5,13 \cdot 10^{-3} \text{ m}^3$$

$$b) \frac{V_0}{T_0} = \frac{V_2}{T_2} \rightarrow V_2 = \frac{8,61 \cdot 10^{-4}}{4,2} \cdot 293 = 6,01 \cdot 10^{-2} \text{ m}^3$$

56.

$$T_1 = 5^\circ \text{C} = 278,15 K \quad T_2 = 25^\circ \text{C} = 298,15 K$$

$$V = 15 \text{ cm}^3 = 15 \cdot 10^{-6} \text{ m}^3$$

$$P_{40} = P_{\text{atm}} + \rho gh = 1,013 \cdot 10^5 + 1000 \cdot 9,81 \cdot 40$$

$$P_{40} = 493700 \text{ Pa}$$

$$\frac{P_{40} V_{40}}{T_{40}} = \frac{P_s V_s}{T_s} \rightarrow \frac{493700 \cdot 15 \cdot 10^{-6}}{278,15} = \frac{1,013 \cdot 10^5 \cdot V_s}{298,15}$$

$$V_s = 78,36 \text{ cm}^3$$

$$53, V = GL = 6 \cdot 10^{-3} \text{ m}^3 \quad m = 10 \text{ g}$$

$$T_0 = 25 \text{ K} \quad T_3 = 293 \text{ K} \quad P = 1 \text{ atm}$$

$$n = \frac{m}{M} \quad M_{\text{He}} = 4 \text{ g/mol}$$

$$n = \frac{10}{4} = 2,5 \text{ mol}$$

$$V = \frac{nRT}{P} \rightarrow \frac{2,5 \cdot 8,314 \cdot 25}{1,013 \cdot 10^5} = 5,13 \text{ L}$$

Asumo que el helio evaporado ocupa todo el recipiente

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \rightarrow P_2 = \frac{P_1 V_1 T_2}{V_2 T_1} = \frac{1 \cdot 5,13 \cdot 293}{6 \cdot 25}$$

$$P_2 = 10,02 \text{ atm}$$

$$54, T_0 = 20^\circ \text{C} = 293,15 \quad T_1 = 50^\circ \text{C} = 323,15 \text{ K}$$

$$P_0 = 2 \cdot 10^5 \text{ Pa}$$

a) A Ucte

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \rightarrow \frac{2 \cdot 10^5}{293,15} = \frac{P_2}{323,15} \rightarrow P_2 = 2,20 \cdot 10^5 \text{ Pa}$$

$$b) \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad V_2 = 1,1 V_1$$

$$P_2 = \frac{323,15 \cdot 2 \cdot 10^5}{293,15 \cdot 1,1} = 200424,9 \text{ Pa}$$

Teoría Cinética de los gases

$$v_{cm} = v_{rms}$$

59. $V = 1 \text{ L}$ $P = 10^5 \text{ Pa}$ $n = 1 \text{ mol}$

$$a) v_{cm} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

$$pV = nRT \quad T = \frac{10^5 \cdot 10^{-3}}{1 \cdot 8,314} = 121,84 \text{ K}$$

$$v_{cm} = \sqrt{\frac{3 \cdot 1013}{40 \cdot 10^{-3}}} = 275,64 \text{ m s}^{-1}$$

b)

$$v_{cm} = \sqrt{\frac{3 \cdot 1013}{4 \cdot 10^{-3}}} = 871,64 \text{ m s}^{-1}$$

60. $T = 273,15 \text{ K}$ $pV = nRT$

$$n = \frac{1 \cdot 1}{0,08206 \cdot 273,15} = 0,0446 \text{ mol}$$

$$K_{tr} = \frac{3}{2} nRT = \frac{3}{2} NkT \rightarrow \text{Para todas las moléculas}$$

$$K_{tr} = \frac{3}{2} \cdot 0,0446 \cdot 8,314 \cdot 273,15 = 151,95 \text{ J}$$

